

Design Rulebook

Project Goals

Generate 3D printable electromagnetic circuits in OpenSCAD. Circuits will consist of a protopasta conductive PLA trace layer and an insulating layer that separates the traces.

Physical Laws

- Conductive trace layer resistivity (ρ): 15 $\Omega \cdot \text{cm}$
- Resistance: $R = \frac{\rho A_c}{L}$
- A_c = cross-sectional area of material
- L = length of material

Design Principles

- Conductive trace width: 2mm
- Conductive trace spacing: 2mm
- Conductive trace height: 2mm
- Minimum Corner Angle: 45 degrees
- Insulating layer thickness: 2mm
- Insulating layer and Conductive Trace layer are generated separately, not as a single OpenSCAD project
- The insulating layer should provide 2mm of clearance on each trace boundary
 - i.e., the trace footprint outline is 25.4 mm by 25.4 mm; the insulating layer footprint should be 29.4 mm by 29.4 mm, matching the general shape of the trace footprint outline.

Optimization Goals

- Use the length and routing needed to use the minimum amount of materials, while accounting for the Design Principles and Physical Laws

Style Preferences

- The insulating layer serves as a flat platform for conductive traces to be printed on
- Only create walls in the insulating layer if necessary to isolate distinct trace planes
- Prefer serpentine-based resistors
- Prefer interdigitated-based capacitors